

Deterministic Mathematical Modeling and Sizing Framework for Distributed Clean Energy: Solar Photovoltaics, Battery Storage, and Electric Vehicle Supply Equipment

A First-Principles Computational Architecture for Zero-Database, Side-Effect-Free Energy Systems Simulation

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Primary Anchor: <https://www.powelab.org/methodology> • **Interactive Tools:** <https://www.powelab.org>

ABSTRACT & EXECUTIVE SUMMARY

Legacy consumer and commercial clean energy planning tools are predominantly built as lead-generation aggregators, utilizing stochastic heuristics, opaque spreadsheet regressions, or centralized database lookups. Such architectures suffer from unverifiable calculations, privacy risks, and severe physical inaccuracies under edge environmental and electrical operating conditions. This technical specification establishes a unified, 100% deterministic mathematical framework for distributed clean energy planning spanning solar photovoltaics (PV), electrochemical battery energy storage systems (BESS), and electric vehicle supply equipment (EVSE). Built entirely on pure, side-effect-free TypeScript engines executing locally in the browser, the architecture eliminates database dependencies while resolving coupled physical phenomena: (1) Perez anisotropic sky diffuse transposition and sub-zero open-circuit voltage (V_{oc}) expansion under NEC 690.7; (2) modified Peukert electrochemical capacity derating with continuous inverter standby tare loss dissipation (P_{tare}) and IEEE 485 depth-of-discharge constraints; (3) continuous-duty thermal ampacity sizing under NFPA 70 (NEC 625.42 and 110.14(C)) alongside Level 1 vs. Level 2 rectifier efficiency divergence; and (4) inductive motor starting Locked Rotor Amperage (LRA) sub-transient reactance (X_d'') voltage dip envelopes. All derivations, governing closed-form equations, boundary conditions, and test vectors are made fully open-access.

Keywords: Deterministic energy modeling, pure functions, solar PV transposition, sub-zero Voc expansion, Peukert's law, inverter tare losses, continuous-duty EVSE, NEC 2023, IEEE 1547, IEEE 485, open science.

1. Introduction & The Deterministic Paradigm

The decentralized transition toward electrified residential and light-commercial energy infrastructure requires tightly coupled sizing across four domains: onsite solar PV generation, chemical battery storage, controllable EV charging loads, and thermal heat pumps.

Historically, web-based planning calculators have abstracted these engineering problems using static "rules of thumb" (e.g., assuming battery runtime is simply capacity divided by load, or that solar production is constant across tilt and ambient temperatures). Such simplifications cause critical engineering failures: undersized inverter DC inputs that fail during cold snaps due to high open-circuit voltage (V_{oc}), severely underperforming battery backup systems caused by unmodeled Peukert capacity derating and inverter idle tare draw, and

dangerous terminal thermal runaway in EVSE installations wired with improper conductor insulation ratings.

PowerLab establishes an open, deterministic paradigm: every calculation is executed by a pure mathematical engine f : **Input** $\rightarrow$ **Result** where output is invariant given identical physical inputs, with zero database dependencies, telemetry overhead, or algorithmic obfuscation.

2. Solar Photovoltaic DC Boundary Modeling

2.1 Perez Anisotropic Transposition

Accurately predicting solar irradiance on tilted arrays requires transposing global horizontal irradiance (GHI , I_{gh}) into plane-of-array irradiance (POA , I_{poa}). PowerLab incorporates the Perez anisotropic sky diffuse model, resolving circumsolar, horizon brightening, and isotropic diffuse components:

$$I_{poa} = I_{bn} \cos \theta + I_{dh} \left[(1 - F_1) \left(\frac{1 + \cos \beta}{2} \right) + F_1 \frac{a}{b} \right] + I_{gm} \left(\frac{1 - \cos \beta}{2} \right)$$

Equation 1: Perez anisotropic transposition model with ground-reflected albedo component ρ .

Where β is the array tilt angle, θ is the angle of incidence, F_1 and F_2 are circumsolar and horizon brightness coefficients, and ρ is the ground albedo reflectance ($\rho \approx 0.20$ for grass/asphalt; $\rho \approx 0.70$ – 0.80 for fresh snow).

2.2 Sub-Zero V_{oc} Expansion Kinetics (NEC 690.7)

Photovoltaic semiconductor bandgaps widen as silicon cell temperature drops, increasing open-circuit voltage (V_{oc}). Failure to calculate peak sub-zero string voltage causes catastrophic overvoltage damage to MPPT charge controllers and string inverters during freezing, sunny winter mornings.

$$V_{\text{string,max}} = N_{\text{series}} \times V_{oc,STC} \times [1 + \alpha_{V_{oc}} \times (T_{\text{min,design}} - 25^{\circ}\text{C})]$$

Equation 2: Sub-zero maximum string voltage under NEC Article 690.7.

Where $\alpha_{V_{oc}}$ is the negative temperature coefficient of open-circuit voltage (typically $-0.26\%/^{\circ}\text{C}$ to $-0.31\%/^{\circ}\text{C}$ for monocrystalline silicon) and $T_{\text{min,design}}$ is the ASHRAE extreme minimum annual dry-bulb temperature. For an array with $V_{oc,STC} = 49.5\text{V}$ and $\alpha_{V_{oc}} = -0.28\%/^{\circ}\text{C}$, an ambient low of -25°C induces a $+14.0\%$ voltage expansion, driving string voltage from 495V to 564.3V .

3. Battery Electrochemical Runtime & Tare Losses

3.1 Modified Peukert Rate-Dependent Capacity

In electrochemical lead-acid and lithium-iron-phosphate (LiFePO₄) chemistries, accessible battery capacity decreases as discharge current I increases due to internal ohmic resistance and mass-transfer polarization. PowerLab implements a modernized Peukert formulation with rate-dependent hour base normalization:

$$t_{\text{effective}} = H \cdot \left(\frac{C_{\text{nominal}}}{I \cdot H} \right)^k$$

Equation 3: Generalized Peukert equation normalized to rated discharge period H (typically 20 hours).

Where k is the Peukert exponent ($k = 1.02$ – 1.05 for modern LiFePO₄; $k = 1.15$ – 1.35 for AGM/Flooded lead-acid). For heavy continuous loads ($> 0.5C$), lead-acid usable capacity collapses by 30% to 45% , while LiFePO₄ maintains over 95% of nominal capacity.

3.2 Inverter Standby Tare Dissipation (P_{tare})

Most planning tools calculate backup runtime simply as $t = E_{\text{bat}}/P_{\text{load}}$. In real off-grid microgrids, inverter self-consumption (quiescent tare loss P_{tare}) and efficiency curve decay at low load factors dominate total system drain:

$$P_{dc,\text{total}} = \frac{P_{\text{load,ac}}}{\eta_{\text{inverter}}(P_{\text{load}})} + P_{\text{tare,standby}}$$

Equation 4: Total DC discharge draw incorporating non-linear inverter efficiency and continuous tare loss.

For a typical 3000W split-phase inverter drawing $P_{tare} = 45\text{W}$ continuously, powering a light 50W emergency medical load results in an actual DC draw of 105W —slashing expected battery autonomy by more than 52% .

4. Continuous-Duty EVSE Sizing & Thermal Physics

4.1 125% Continuous Duty Sizing (NEC 625.42)

Electric vehicle supply equipment operates continuously for hours, creating steady-state Joule heating (I^2R) within conductors, breaker enclosures, and busbars. NFPA 70 defines any load sustained for 3 hours or more as a continuous load, requiring a 125% overcurrent protection multiplier:

$$I_{\text{breaker}} \geq I_{\text{nameplate,continuous}} \times 1.25$$

Equation 5: Branch circuit overcurrent protective device (OCPD) minimum ampacity.

A residential 48A continuous charging load requires a minimum breaker rating of $48\text{A} \times 1.25 = 60\text{A}$.

4.2 Terminal Temperature Limits (NEC 110.14(C))

Conductor sizing cannot be based solely on 90°C insulation tables (e.g., THHN). Under NEC 110.14(C), equipment terminals rated 100A or less are capped at 60°C or 75°C ampacity ratings. Furthermore, non-metallic sheathed cable (NM-B / Romex) is strictly capped at 60°C ampacity under NEC 334.80:

CRITICAL CODE RESTRICTION: 6 AWG NM-B VIOLATION

6 AWG copper in NM-B cable is limited to 55A (60°C column). Installing 6 AWG Romex on a 60A breaker for a 48A continuous EVSE violates NEC 334.80 and creates severe thermal degradation risk. Code compliance mandates THHN in conduit (rated 65A at 75°C) or 4 AWG NM-B (rated 70A at 60°C).

4.3 Level 1 vs Level 2 Rectification Efficiency

Charging an EV on standard 120V Level 1 yields only 72% to 78% AC-to-DC efficiency due to constant onboard BMS, coolant pump, and computer parasitic overhead ($\sim 250\text{W}$ – 350W). On 240V Level 2, charging duration is shortened by

4× to 6×, elevating net efficiency to 89%–94% and saving 15% to 20% in gross electrical energy per charge cycle.

5. Motor Inrush Dynamics & Generator Peak Sizing

Standby generator sizing fails when relying strictly on steady-state continuous wattage. Single-phase induction motors (well pumps, HVAC compressors, sump pumps) exhibit Locked Rotor Amperage (LRA) drawing 4.5× to 7.0× Full Load Amps (FLA) during across-the-line starting (0.1s–0.4s).

$$S_{\text{surge,kVA}} = \frac{V_{\text{line}} \times I_{\text{LRA}}}{1000} \leq S_{\text{alt,peak}} \times (1 - \Delta V_{\text{dip,max}})$$

Equation 6: Motor starting apparent power and generator sub-transient reactance headroom.

PowerLab applies non-coincident load stacking: generator capacity is sized for continuous steady-state loads plus the single largest simultaneous inductive starting surge, adjusted for fuel derating (Gasoline = 1.00; Propane = 0.90; Natural Gas = 0.80) and altitude (3.5% derating per 1,000ft elevation).

6. Deterministic Engine TypeScript Architecture

All calculations are formalized as pure functions adhering to a standardized result envelope containing transparent assumptions, provenance, and warnings:

```
export interface CalculationResult<T> {
  formulaVersion: string;
  result: T;
  assumptions: Array<{
    key: string;
    value: number | string;
    unit?: string;
    provenance: "measured" | "preset" | "standard";
    description: string;
  }>;
  warnings: Array<{
    code: string;
    severity: "info" | "caution";
    message: string;
  }>;
  qualityLabel: "specific-inputs" | "preset-assisted";
}
```

7. Standards Cited & References

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